CHAPTER 11

Matrix sequences and Markov chains

Sequences $U_{n+1} = AU_n + B$, probabilistic graphs, invariant distributions

Objectives. Study a sequence of column matrices defined by $U_{n+1} = AU_n + B$; model a random evolution with two or three states by a probabilistic graph and a transition matrix; compute the distribution after n steps; find an invariant distribution; study convergence in the two-state case.

Prerequisites. Chapters 9 and 10; conditional probability and the law of total probability; arithmetico-geometric sequences; limit of a geometric sequence.

Fluency

Fluency · No calculator, five minutes.

1. Let $u_{n+1} = 0.5u_n + 2$. Find the real number ℓ such that $\ell = 0.5\ell + 2$.
2. Give the limit of 0.8^n as n tends to $+\infty$.
3. In a tree, $P(A) = 0.3$, $P_A(B) = 0.6$, $P_{\overline{A}}(B) = 0.2$. Compute $P(B)$.
4. Solve the system $0.2a = 0.4b$ and $a + b = 1$.
5. Compute $\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}0.8 & 0.2\\ 0.4 & 0.6\end{pmatrix}$.
6. What is the sum of the entries of each row of the matrix in question 5?

Solution · Fluency.

1. $\ell = 4$.
2. 0.
3. $0.3 \times 0.6 + 0.7 \times 0.2 = 0.32$.
4. $a = 2b$, so $3b = 1$: $a = \frac{2}{3}$ and $b = \frac{1}{3}$.
5. $\begin{pmatrix}0.8 & 0.2\end{pmatrix}$.
6. 1 for each row.

Introductory activity

Activity 1 · Fine weather, rain 40 min . In a simplified model (fictitious data), the weather on a given day is either fine (S) or rainy (P), and depends only on the weather the day before: after a fine day, it is fine with probability 0.8; after a rainy day, it is fine with probability 0.4. Day 0 is fine. Let s_n and p_n be the probabilities that day n is fine or rainy.

1. Represent the situation by a diagram: two vertices S and P, an arrow for each possible change, carrying its probability.
2. Using a tree, compute s_1, p_1, then s_2, p_2.
3. Justify that $s_{n+1} = 0.8s_n + 0.4p_n$ and $p_{n+1} = 0.2s_n + 0.6p_n$.
4. Let $\pi_n = \begin{pmatrix} s_n & p_n \end{pmatrix}$ and $T = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$. Check that $\pi_{n+1} = \pi_n T$. Compute π_{10} with a calculator. Start again with a rainy day 0. What do you notice?
5. Find the distribution $\pi = \begin{pmatrix} a & b \end{pmatrix}$, with $a + b = 1$, that satisfies $\pi T = \pi$. Compare it with the observations of question 4.

Observing is not proving: the properties conjectured here are established in the course.

Solution · Activity.

1. Arrows $S \to S$ (0.8), $S \to P$ (0.2), $P \to S$ (0.4), $P \to P$ (0.6).
2. $s_1 = 0.8$, $p_1 = 0.2$; $s_2 = 0.8 \times 0.8 + 0.2 \times 0.4 = 0.72$ and $p_2 = 0.28$.
3. Law of total probability with the partition "fine on day n", "rain on day n".
4. $\pi_n T = \begin{pmatrix} 0.8s_n + 0.4p_n & 0.2s_n + 0.6p_n \end{pmatrix} = \pi_{n+1}$. We find $\pi_{10} \simeq \begin{pmatrix} 0.6667 & 0.3333 \end{pmatrix}$, and $\pi_{10} \simeq \begin{pmatrix} 0.6666 & 0.3334 \end{pmatrix}$ starting from a rainy day. The distribution seems to settle at $\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix}$, whatever the start.
5. $0.8a + 0.4b = a$ is equivalent to $0.2a = 0.4b$, that is $a = 2b$; with $a + b = 1$: $\pi = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix}$, the limit observed.

Teacher note · Running the activity. The numerical computations of question 4 suggest convergence; the proof, in the two-state case, uses an arithmetico-geometric sequence (course). Draw attention to writing π_n as a row and to the order of the product $\pi_n T$.

Course notes

Matrix sequences

Property · Sequence $U_{n+1} = AU_n + B$. Let A be a square matrix of order p, B a column matrix and (U_n) a sequence of column matrices such that $U_{n+1} = AU_n + B$ for every n. If $I_p - A$ is invertible, the matrix $C = (I_p - A)^{-1}B$ satisfies $C = AC + B$, and for every natural number n:

$$U_n = A^n (U_0 - C) + C.$$

Proof. $(I_p - A)C = B$ can be written $C = AC + B$. Subtracting: $U_{n+1} - C = A(U_n - C)$. The sequence $V_n = U_n - C$ satisfies $V_{n+1} = AV_n$, so $V_n = A^n V_0$ (induction), and $U_n = A^n(U_0 - C) + C$. □

Key point · Behaviour at infinity. If all the entries of A^n tend to 0, then U_n converges to C (entry by entry). This behaviour can be observed with a calculator; it is taken for granted when the statement says so.

Markov chains

Definition · Markov chain with two or three states. A system is observed at times $0, 1, 2, \ldots$; at each time, it is in one of the states $1, \ldots, k$ ($k = 2$ or 3). Let X_n be the state at time n. We say that (X_n) is a (homogeneous) *Markov chain* if the probability of moving from state i to state j depends neither on n nor on the earlier states: for all i, j and n such that $P(X_n = i) \neq 0$,

$$P_{(X_n=i)}(X_{n+1} = j) = t_{ij}.$$

The numbers t_{ij} are the *transition probabilities*.

Definition · Probabilistic graph, transition matrix. The chain is represented by a *probabilistic graph*: a weighted directed graph whose vertices are the states, with an arc from i to j, of weight t_{ij}, when $t_{ij} > 0$. The *transition matrix* is the square matrix $T = (t_{ij})$. Its entries are nonnegative and the sum of each row is 1.

Definition · Distribution. The *distribution* at time n is the row matrix $\pi_n = \begin{pmatrix} P(X_n = 1) & \cdots & P(X_n = k) \end{pmatrix}$; π_0 is the initial distribution.

Property · Evolution of the distribution. For every natural number n: $\pi_{n+1} = \pi_n T$ and $\pi_n = \pi_0 T^n$.

Proof. By the law of total probability, with the partition $(X_n = 1), \ldots, (X_n = k)$ (events of probability zero are left out, as they contribute 0):

$$P(X_{n+1} = j) = \sum_{i=1}^{k} P(X_n = i)\, t_{ij},$$

which is entry j of $\pi_n T$. The second formula follows by induction. □

Property · Transition in n steps. If $P(X_0 = i) = 1$, then for every state j, $P(X_n = j)$ is the entry in row i and column j of T^n. In other words, the probability of moving from state i to state j in n transitions is $(T^n)_{ij}$.

Proof. The initial distribution is the row π_0 with a 1 in position i and 0s elsewhere. Then $\pi_n = \pi_0 T^n$ is row i of T^n. □

Definition · Invariant distribution. A distribution π (a row of nonnegative entries with sum 1) is *invariant* if $\pi T = \pi$. If $\pi_0 = \pi$, then $\pi_n = \pi$ for every n.

Property · Two-state case. Let $T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $0 \leq a \leq 1$, $0 \leq b \leq 1$ and $0 < a + b < 2$. The chain has a unique invariant distribution $\pi = \begin{pmatrix} \frac{b}{a+b} & \frac{a}{a+b} \end{pmatrix}$, and whatever the initial distribution, π_n converges to π.

Proof. Write $\pi_n = \begin{pmatrix} x_n & 1-x_n \end{pmatrix}$. Then $x_{n+1} = (1-a)x_n + b(1-x_n) = (1-a-b)x_n + b$. This arithmetico-geometric sequence has fixed point $\ell = \frac{b}{a+b}$, and $x_n - \ell = (1-a-b)^n(x_0 - \ell)$. Since $-1 < 1-a-b < 1$, $(1-a-b)^n \to 0$ and $x_n \to \ell$. An invariant distribution satisfies $x = (1-a-b)x + b$, so $x = \ell$: it is unique. □

Method · Three states: invariant distribution. Solve the system formed by $\pi T = \pi$ and the condition $a + b + c = 1$; one of the three equations of $\pi T = \pi$ follows from the other two. Possible convergence of π_n is observed with a calculator.

Watch out. The distribution is a row and we multiply on the right by T: $\pi_{n+1} = \pi_n T$, not $T\pi_n$. It is the rows of T, not its columns, that add up to 1.

Exercises

Exercise 11.1 · From the graph to the matrix. A probabilistic graph has three vertices 1, 2, 3 and the following arcs: $1 \to 1$ ($\frac{1}{2}$), $1 \to 2$ ($\frac{1}{4}$), $1 \to 3$ ($\frac{1}{4}$), $2 \to 1$ ($\frac{1}{2}$), $2 \to 3$ ($\frac{1}{2}$), $3 \to 1$ ($\frac{1}{4}$), $3 \to 2$ ($\frac{1}{4}$), $3 \to 3$ ($\frac{1}{2}$).

1. Write the transition matrix T and check the row sums.
2. Starting from state 1, compute the distribution after two transitions.

Solution.

1. Each row adds up to 1:
$$T = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}.$$
2. $\pi_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$, then $\pi_2 = \pi_1 T = \begin{pmatrix} \frac{1}{4}+\frac{1}{8}+\frac{1}{16} & \frac{1}{8}+0+\frac{1}{16} & \frac{1}{8}+\frac{1}{8}+\frac{1}{8} \end{pmatrix} = \begin{pmatrix} \frac{7}{16} & \frac{3}{16} & \frac{3}{8} \end{pmatrix}$.

Exercise 11.2 · Two network operators. Each month, 10% of the customers of operator A move to B and 20% of the customers of B move to A; the others stay. At the start, each operator has half of the customers. Let x_n be the share of A after n months.

1. Write the transition matrix and find the invariant distribution.
2. Prove that $x_{n+1} = 0.7x_n + 0.2$, then that $x_n = \frac{2}{3} - \frac{1}{6} \times 0.7^n$.
3. Find the smallest n such that $x_n > 0.66$.

Solution.

1. $T = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}$. With $a = 0.1$ and $b = 0.2$, the course gives $\pi = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix}$.
2. $x_{n+1} = 0.9x_n + 0.2(1 - x_n) = 0.7x_n + 0.2$. The fixed point is $\frac{2}{3}$ and $x_n - \frac{2}{3} = 0.7^n \left(\frac{1}{2} - \frac{2}{3}\right) = -\frac{1}{6} \times 0.7^n$.
3. $x_n > 0.66$ is equivalent to $\frac{1}{6} \times 0.7^n < \frac{2}{3} - 0.66 = \frac{1}{150}$, that is $0.7^n < 0.04$, that is $n > \frac{\ln 0.04}{\ln 0.7} \simeq 9.02$. The smallest integer is $n = 10$ ($0.7^9 \simeq 0.0404$ and $0.7^{10} \simeq 0.0282$).

Exercise 11.3 · A three-state walk. A particle moves on three squares in a row, 1, 2, 3. From square 2, it goes to 1 or to 3 with probability $\frac{1}{4}$ each and stays on 2 with probability $\frac{1}{2}$. From an end square, it stays where it is or goes to 2, with probability $\frac{1}{2}$ each.

1. Write the transition matrix T.
2. Check that $\begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$ is an invariant distribution.
3. Prove that it is the only one.

Solution.

1. In the order of the squares 1, 2, 3:
$$T = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$
2. $\frac{1}{4}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{4}=\frac{1}{4}$; $\frac{1}{4}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{2}+\frac{1}{4}\times\frac{1}{2}=\frac{1}{2}$; $\frac{1}{2}\times\frac{1}{4}+\frac{1}{4}\times\frac{1}{2}=\frac{1}{4}$.
3. With $\pi = \begin{pmatrix} a & b & c \end{pmatrix}$, the first equation gives $\frac{a}{2}+\frac{b}{4}=a$, that is $b=2a$; the third gives likewise $b=2c$. With $a+b+c=1$: $a+2a+a=1$, hence $a=c=\frac{1}{4}$ and $b=\frac{1}{2}$.

Exercise 11.4 · Two tanks. The amount of water (in thousands of litres) in two connected tanks is modelled by $U_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix}$ with
$$U_{n+1} = AU_n + B, \qquad A = \begin{pmatrix} 0.6 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}, \qquad B = \begin{pmatrix} 10 \\ 5 \end{pmatrix}.$$
1. Find the column C such that $C = AC + B$.
2. Express U_n in terms of A^n, U_0 and C.
3. We take for granted that the entries of A^n tend to 0. Find the limit of U_n. Does it depend on U_0?

Solution.

1. $(I_2 - A)C = B$ with $I_2 - A = \begin{pmatrix} 0.4 & -0.2 \\ -0.1 & 0.3 \end{pmatrix}$, whose determinant is $0.12 - 0.02 = 0.1$. So $C = \frac{1}{0.1}\begin{pmatrix} 0.3 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}\begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 40 \\ 30 \end{pmatrix}$. Check: $0.6\times 40 + 0.2\times 30 + 10 = 40$ and $0.1\times 40 + 0.7\times 30 + 5 = 30$.
2. $U_n = A^n(U_0 - C) + C$.
3. $U_n \to C = \begin{pmatrix} 40 \\ 30 \end{pmatrix}$, whatever the value of U_0.

Exercise 11.5 · Transitions in three steps. Each day, a user of an app is active (1) or inactive (2), with transition matrix $T = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$.

1. Compute T^2 and T^3.
2. A user is active on day 0. What is the probability that they are inactive on day 3?
3. Find the invariant distribution.

Solution.

1. $T^2 = \begin{pmatrix} 0.64 & 0.36 \\ 0.6 & 0.4 \end{pmatrix}$ and $T^3 = \begin{pmatrix} 0.628 & 0.372 \\ 0.62 & 0.38 \end{pmatrix}$.

2. It is the entry $(1,2)$ of T^3: 0.372.
3. $a = 0.3$ and $b = 0.5$: $\pi = \left(\frac{0.5}{0.8} \quad \frac{0.3}{0.8}\right) = \left(\frac{5}{8} \quad \frac{3}{8}\right)$.

Exercise 11.6 · A no-claims bonus system. Each year an insurer classes its customers in three levels: bonus (1), standard (2), penalty (3). The annual transition matrix is

$$T = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.6 & 0.2 & 0.2 \\ 0 & 0.6 & 0.4 \end{pmatrix}.$$

1. Draw the probabilistic graph.
2. Find the invariant distribution.
3. A customer is at the standard level. What is the probability of being at the bonus level two years later?

Solution.

1. Arcs $1 \to 1$ (0.8), $1 \to 2$ (0.2), $2 \to 1$ (0.6), $2 \to 2$ (0.2), $2 \to 3$ (0.2), $3 \to 2$ (0.6), $3 \to 3$ (0.4).
2. With $\pi = \begin{pmatrix} a & b & c \end{pmatrix}$: $0.8a + 0.6b = a$ gives $b = \frac{a}{3}$; $0.2b + 0.4c = c$ gives $c = \frac{b}{3} = \frac{a}{9}$. Then $a + \frac{a}{3} + \frac{a}{9} = 1$, that is $a = \frac{9}{13}$, $b = \frac{3}{13}$, $c = \frac{1}{13}$.
3. Entry $(2,1)$ of T^2: $0.6 \times 0.8 + 0.2 \times 0.6 + 0.2 \times 0 = 0.6$.

Exercise 11.7 · Powers of the two-state transition matrix. Let $T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $a + b \neq 0$. Let $\lambda = 1 - a - b$, $Q = \frac{1}{a+b}\begin{pmatrix} b & a \\ b & a \end{pmatrix}$ and $R = \frac{1}{a+b}\begin{pmatrix} a & -a \\ -b & b \end{pmatrix}$.

1. Check that $Q + \lambda R = T$, $Q^2 = Q$, $R^2 = R$ and $QR = RQ = 0$.
2. Deduce by induction that $T^n = Q + \lambda^n R$ for every $n \geq 1$.
3. Assume $0 < a + b < 2$. Find the limit of T^n and interpret its rows.

Solution.

1. The entry $(1,1)$ of $Q + \lambda R$ is $\frac{b + (1-a-b)a}{a+b} = \frac{(a+b) - a(a+b)}{a+b} = 1 - a$; the others are checked in the same way. For Q^2: the entry $(1,1)$ is $\frac{b^2 + ab}{(a+b)^2} = \frac{b}{a+b}$, and so on. QR has entry $(1,1)$ equal to $\frac{ba - ab}{(a+b)^2} = 0$; all the entries of QR and RQ are zero.
2. True for $n = 1$. If $T^n = Q + \lambda^n R$, then $T^{n+1} = (Q + \lambda^n R)(Q + \lambda R) = Q^2 + \lambda QR + \lambda^n RQ + \lambda^{n+1} R^2 = Q + \lambda^{n+1} R$.
3. $|\lambda| < 1$, so $\lambda^n \to 0$ and $T^n \to Q$. Both rows of Q equal the invariant distribution: in the long run, the starting state is "forgotten".

Problem

Problem · Ehrenfest's urn model. Two urns U and V contain two balls in total. At each step, one of the two balls is chosen at random (with equal probability) and moved to the other urn. Let X_n be the number of balls in U after n steps; the states are 0, 1, 2. At the start, U contains both balls.

1. Justify that (X_n) is a Markov chain, draw its probabilistic graph and write its transition matrix T (states in the order 0, 1, 2).
2. Find the unique invariant distribution π. Compare it with the distribution of the number of balls in U when each ball is placed at random in one of the two urns, independently.
3. Compute π_1, π_2, π_3, π_4. Does the sequence (π_n) converge?
4. Prove that for every $k \geq 1$, $\pi_{2k} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$ and $\pi_{2k+1} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$, then that $\frac{1}{2}(\pi_{2k} + \pi_{2k+1}) = \pi$.
5. Explain, without computation, why X_n has the same parity as n.

Solution · Problem.

1. The next step depends only on the current number of balls in U. From state 0 or 2, the chain moves to 1 for certain; from state 1, it moves to 0 or to 2 with probability $\frac{1}{2}$ each.

$$T = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix}.$$

2. $\pi T = \pi$ with $\pi = \begin{pmatrix} a & b & c \end{pmatrix}$ gives $\frac{b}{2} = a$, $a + c = b$, $\frac{b}{2} = c$. With $a + b + c = 1$: $\pi = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$. This is the binomial distribution with parameters 2 and $\frac{1}{2}$.
3. $\pi_0 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$, $\pi_1 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$, $\pi_2 = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$, $\pi_3 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$, $\pi_4 = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$. The sequence alternates between two values: it does not converge.
4. $\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} T = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 & 0 \end{pmatrix} T = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$; an induction on k concludes. The half-sum is $\begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix} = \pi$.
5. Each step changes the number of balls in U by 1, so it changes its parity. At the start $X_0 = 2$ is even.

Teacher note · Status of the problem. The Ehrenfest model is a "possible problem" of the syllabus. It shows that an invariant distribution can exist without π_n converging: the condition $0 < a + b < 2$ of the two-state case has an analogue for three states, beyond the syllabus (aperiodic chain). Extension: simulate the chain in Python with N balls and observe the frequencies of the states over a long run.